

CDR1 and CDR2 encoded by germline in V

CDR3 by DJ and junctional for VDJ B and J + VJ junctional for a

α chain
Chr 14

V

n=~45

J

n=~50

C

β chain
chr 7

V

n=~50

D

n=2

J

n=12

C

junctional diversity
(non-germline)

largest contributor to diversity

P nucleotides (template after hairpin opening by Artemis)

N nucleotides (then nontemplate followed by DNA ligation) (20 by TdT) - more common in B chain

combinatorial diversity
(germline)

1 to 3 million upper limit)

10^{15} - 10^{20} unique TCR combinations